Biconnected Graph Triangulation Algorithms

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December 4, 2025

Complete Algorithms

StartTriangulation

Algorithm 1 StartTriangulation

procedure StartTriangulation(G, k)
    Label faces as $F_1, F_2, \ldots, F_k$ arbitrarily
    Initialize global chord set S
    $T \leftarrow \emptyset$ ▷ Empty partial triangulation
    FindAllTriangulationsCycle($F, T, 0$)
end procedure

FindSafeRootTriangulation

Algorithm 2 FindSafeRootTriangulation

procedure FindSafeRootTriangulation(F, index)
    startIndex $\leftarrow n - 1$
    endIndex $\leftarrow 0$
    while startIndex $>$ endIndex $+ 1$ **do**
        if chord $(v_{\text{startIndex}}, v_{\text{endIndex}})$ exists in S **then**
            startIndex $\leftarrow$ startIndex $- 1$
        else
            endIndex $\leftarrow$ endIndex $+ 1$
        end if
    end while
    $v_r \leftarrow v_{\text{endIndex}}$ ▷ Root vertex for safe triangulation
    Build fan triangulation T_r from v_r as root vertex
    return T_r
end procedure

Time Complexity: $\mathcal{O}(n)$ where n is the number of vertices in the face.

FindAllTriangulationsCycle

Algorithm 3 FindAllTriangulationsCycle

procedure $\textsc{FindAllTriangulationsCycle}(F, T, i)$
    $T_i \leftarrow \textsc{FindSafeRootTriangulation}(F, i)$ ▷ $\mathcal{O}(n)$
    if F_i is the last face **then** ▷ All faces triangulated
        output $(T \cup T_i)$
    else
        $\textsc{FindAllTriangulationsCycle}(F, T \cup T_i, i + 1)$
    end if
    for $j = n - 1$ **down to** 3 **do**
        $\textsc{FindAllChildTriangulationsCycle}(T_i(v_r, v_j), T, i)$
    end for
end procedure

FindAllChildTriangulationsCycle

Algorithm 4 FindAllChildTriangulationsCycle

procedure $\textsc{FindAllChildTriangulationsCycle}(T_i, T, i)$
    if F_i is the last face **then** ▷ All faces triangulated
        output $(T \cup T_i)$
    else
        $\textsc{FindAllTriangulationsCycle}(F, T \cup T_i, i + 1)$
    end if
    Let $(v_b, v_{b'})$ be the leftmost generating chord of T_i
    for j from b to $n - 1$ **do**
        if (v_r, v_j) is a chord of T_i **or** $oppositePair(v_r, v_j) \notin S$ **then**
            $\textsc{FindAllChildTriangulationsCycle}(T_i(v_r, v_j), T, i)$
        end if
    end for
end procedure